MESA · 6 × 90 MIN

From AI Foundations to AI Agents

A hands-on practitioner series for building intuition, prompting skill, grounded AI systems and action-taking agents.

DIVIJ BAJAJ

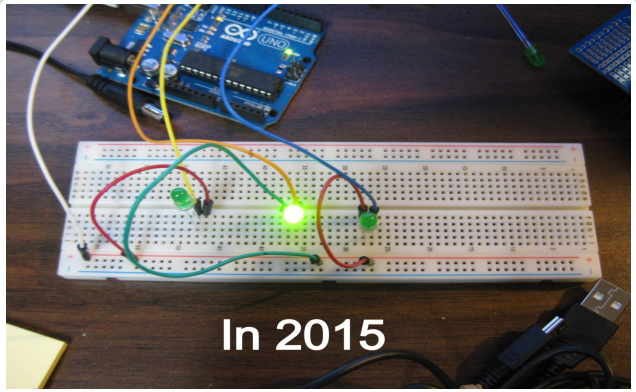
Microsoft Senior Data & Applied Scientist · Founder, Thinklytics AI

UNDERSTAND → BUILD → TEST

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MEET YOUR FACILITATOR

Builder first. Teacher second.



Embedded systems

Built with Arduino, Raspberry Pi and sensors—the point where data became real.



Business + data

MBA in Data Science & Business Analytics; Gold Medallist, Symbiosis.



AI at scale

Built M365 AI Copilot for anomaly detection at Microsoft; now enabling enterprise teams.

THE LEARNING JOURNEY

Six sessions. One connected system.



Each session adds one layer: model → instruction → knowledge → product → action → autonomy.

HOW WE WILL WORK

Less watching. More doing.

01 CONCEPT

Build a mental model

02 DEMO

See the idea behave

03 BUILD

Try it on a real task

04 TEST

Break it, score it, improve it

Working norm: use dummy, public or sanitised data. AI drafts; humans decide.

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PROGRAM OUTCOMES

What participants will be able to do

- Explain how modern AI systems learn, generate and fail—without jargon.
- Write, evaluate and iterate robust prompts across ChatGPT, Claude and Gemini.
- Describe embeddings, vector search and RAG—and make sensible cost choices.
- Build and test a grounded Nykaa product-discovery assistant.
- Choose between an API, webhook, workflow, chatbot and agent.
- Design an agent loop with instructions, memory, tools and measurable success.

SESSION 1 · 90 MIN

How AI Works

Demystify the model—from training data to one token at a time—and learn where confidence becomes risk.

SESSION FLOW

- Hook + live prediction
- AI / ML / Generative AI
- Train + generate
- Limits + hallucinations
- Teachable Machine

HANDS-ON

Train a tiny image classifier

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SESSION 1 · OPENING

AI is not Google. So what is it?

SEARCH ENGINE

Finds a page that may contain the answer

≠

GENERATIVE AI

Generates a response from learned patterns + your context

Opening challenge: Which one can sound certain without having a source?

SESSION 1 · CONCEPT MAP

AI, ML, Generative AI—and where LLMs fit

AI

Artificial Intelligence

The umbrella: machines performing tasks associated with human intelligence.

GEN

Generative AI

Creates new text, images, audio, video or code from learned patterns.

ML

Machine Learning

Learns patterns from examples rather than following only handwritten rules.

LLM

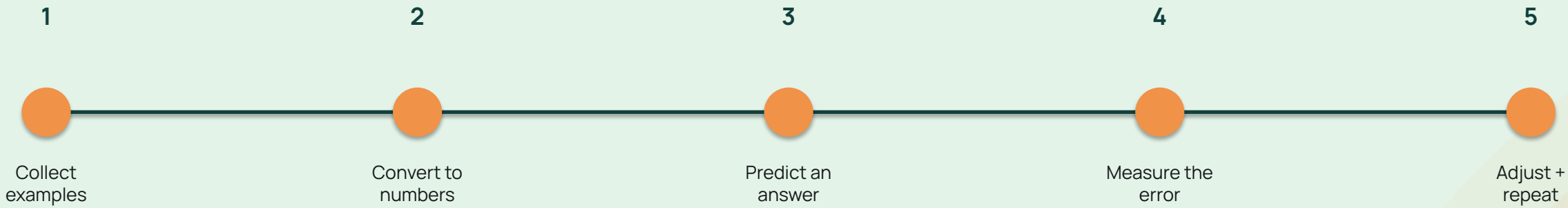
Large Language Models

A family of generative AI models trained to understand and generate language token by token.

Predictive AI asks: “What happens next?” · Generative AI asks: “What can I create next?”

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How a model learns



Training learns the pattern. Prompting supplies the situation.

How an LLM responds in five steps

1 TOKENIZE

Break input into pieces

2 EMBED

Turn meaning into numbers

3 SELF-ATTENTION

Focus on relevant context

4 PREDICT

Rank the next token

5 GENERATE

Repeat until complete

It is a prediction engine—not a database, oracle or decision-maker.

Three knobs that change the answer

1 TOKENS

The pieces the model reads and writes.

What did my prompt cost?

2 CONTEXT WINDOW

Everything the model can consider at once.

What did it forget or miss?

3 TEMPERATURE

How much randomness it uses when selecting the next token.

Do I want precision or variety?

Low temperature for extraction and policy. Higher temperature for ideation and creative options.

Fluency is not the same as truth

CAN

Draft, transform, classify, summarise, compare, explain and brainstorm.

CANNOT

Know private or current facts it has not been given; guarantee accuracy; own a decision.

HALLUCINATES WHEN

Evidence is missing, the question assumes a false premise, or the prompt rewards an answer over honesty.

Design response: give evidence, permit “I don’t know”, ask for sources, and keep a human accountable.

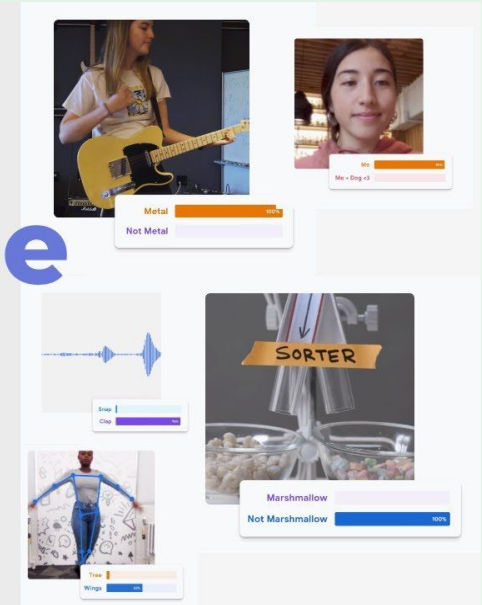
SESSION 1 · HANDS-ON

Train a tiny model with Google Teachable Machine

Google

Teachable Machine

Create Image Data set



01 BUILD

Create 2–3 image classes, capture examples, train and test live.

02 BREAK

Change lighting, angle or background. Observe what the model actually learned.

03 REFLECT

Where could biased examples create a confidently wrong business outcome?

If the answer is wrong, where would you look first?

A model gives two different answers to the same question—and confidently invents a source.

- A Training data
- B Prompt + context
- C Temperature
- D Missing evidence

Big question: When should we trust an AI output—and what evidence earns that trust?

SESSION 2 · 90 MIN

Prompting as a Skill

Turn vague asks into reliable briefs, then use structured reasoning patterns to improve hard tasks.

SESSION FLOW

- Prompt mindset
- 6-Part Formula
- CoT + ToT
- Meta + ReAct
- Evaluate + iterate

HANDS-ON

Prompt lab: one task, five upgrades

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A prompt is a brief—not a search query

SEARCH MINDSET

Use the fewest words and expect the system to find the right page.

BRIEFING MINDSET

State the job, context, boundaries and what good looks like.

The model is not lazy. It is under-briefed.

SESSION 2 · CORE FRAMEWORK

The 6-Part Prompt Formula

1 ROLE

Who is the AI?

2 TASK

What exact job?

3 CONTEXT

What must it know?

4 CONSTRAINTS

What rules apply?

5 FORMAT

What should output look like?

6 EXAMPLE

What does good look like?

Fill more fields only when they help. Precision beats prompt length.

Choose the pattern for the problem

CoT Chain of Thought

Decompose a multi-criterion task; require evidence by criterion; place the conclusion last.

ToT Tree of Thoughts

Generate competing expert lenses, surface disagreement, then converge on one recommendation.

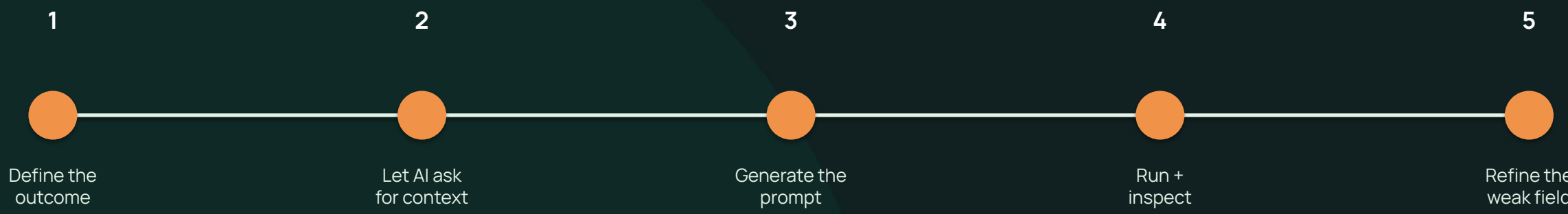
R+A ReAct

Alternate reasoning and action: decide what information or tool is needed, use it, observe, continue.

Use structured outputs—not hidden private reasoning—as the auditable trail.

Meta prompting: ask AI to design the brief

"Interview me one question at a time until you have enough context to create a robust prompt for…"



ChatGPT, Claude and Gemini: adapt, don’t reinvent

1 CHATGPT

Strong general workspace and custom assistants. Use explicit tools, data and output schemas.

2 CLAUDE

Strong long-document work and Artifacts. Use clear XML-like sections for complex context.

3 GEMINI

Strong multimodal and Google ecosystem workflows. Ground requests in attached Google Drive files.

Portable principle: the six fields matter more than brand-specific prompt folklore.

A prompt is only good if the output is consistently good

1 ACCURACY

Is it factually correct?

2 RELEVANCE

Did it answer the real ask?

3 COMPLETENESS

Did it cover all required parts?

4 FORMAT

Can the output be used immediately?

5 SAFETY

Did it respect boundaries?

Test the same prompt on 5–10 representative inputs. Change one thing at a time.

SESSION 2 · HANDS-ON

One task. Five prompt upgrades.

1 ROUND 1

Write the naïve one-line ask

2 ROUND 2

Add the 6-Part Formula

3 ROUND 3

Add a reasoning pattern

4 ROUND 4

Use meta prompting to find gaps

5 ROUND 5

Score, revise and save the winner

Deliverable: one reusable prompt + a five-criterion scorecard.

SESSION 2 · CLOSE

Which prompt would survive a handover?

Your colleague should be able to run it tomorrow—without you explaining what you meant.

SAVE the reusable brief

TEST representative cases

SHARE the scorecard

SESSION 3 · 90 MIN

How AI Understands & Retrieves

Move from token-level meaning to semantic search—and understand why external knowledge changes reliability and cost.

SESSION FLOW

- Embedding intuition
- Similarity search
- RAG flow
- Hosted vs. local
- Token economics

HANDS-ON

Human vector search + cost lab

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Meaning becomes coordinates

“stays all day”	≈	“16-hour transfer-proof”	≠	“banana”
[0.22, 0.51, −0.13, …]		[0.23, 0.49, −0.14, …]		[−0.71, 0.88, 0.04, …]

Similar meaning → nearby points. Different wording can still retrieve the right evidence.

Keyword search matches words. Semantic search matches intent.

KEYWORD SEARCH

Query: “lipstick that lasts all day”

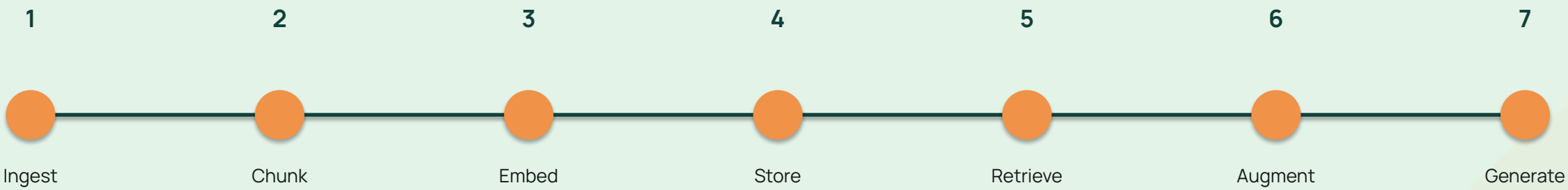
Finds only records containing those words. Misses “16-hour transfer-proof matte.”

SIMILARITY SEARCH

Embeds the query and finds nearby meaning.

Retrieves the right product even when catalogue language differs.

Look it up first. Then answer.



Steps 1–4 prepare the knowledge base. Steps 5–7 happen for every question.

A brilliant new hire who has never seen your files

WHAT THE MODEL HAS

Language fluency, broad public patterns, reasoning ability and general knowledge.

WHAT YOUR BUSINESS NEEDS

Private documents, current facts, permissions, source citations and an explicit path to “I don’t know.”

RAG does not retrain the model. It changes what evidence the model can read before answering.

Hosted or local? Start with the constraints.

1 HOSTED

Fast setup · strongest frontier models · usage-based cost · data leaves your environment under provider terms.

2 LOCAL / SELF-HOSTED

More infrastructure · model control · data residency options · capacity and maintenance become your problem.

3 HYBRID

Sensitive retrieval stays controlled; selected tasks use hosted models under policy and contracts.

Decision factors: sensitivity · latency · quality · scale · skills · total cost · governance.

Cost follows what you send, retrieve and generate

COST PER QUERY

≈ input tokens × input rate + output tokens × output rate

1 **INPUT**

System prompt + conversation + retrieved chunks

2 **OUTPUT**

Generated response and structured data

3 **MULTIPLIER**

Queries per user × users × days

Optimise relevance before length: fewer high-quality chunks often improve both cost and accuracy.

SESSION 3 · HANDS-ON

Be the vector database

- Each group receives product cards with deliberately different wording.
- Map meaning by intuition: arrange cards on a table so similar meanings sit close together.
- Run five shopper queries; retrieve the nearest three cards.
- Compare keyword matches with meaning matches.
- Estimate token cost when retrieving 3, 10 and 30 chunks.

SESSION 3 · CLOSE

Retrieval quality sets the ceiling

The model cannot produce a grounded answer from evidence the retriever never found.

Big question: What content would your organisation trust enough to put behind an AI assistant?

SESSION 4 · 90 MIN

Building RAGs — Nykaa Use Case

Turn the seven-step architecture into a working product-discovery assistant—and learn to debug evidence before prose.

SESSION FLOW

- Frame the shopper need
- Prepare catalogue
- Configure retrieval
- Ground the answer
- Evaluate failures

HANDS-ON

Build a Nykaa assistant in Dify

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A shopper describes a need—not a SKU

> “Something for oily skin that will survive Chennai humidity, under ₹800.”

NORMAL SITE SEARCH

Matches literal words; returns hundreds of popular products.

RAG ASSISTANT

Retrieves products matching the shopper’s intent and filters, then explains why each fits.

The build has two pipelines

A OFFLINE · PREPARE

Catalogue → clean → chunk → embed → vector database

B ONLINE · ANSWER

Question → embed → retrieve → augment → generate + cite

Preparation determines what can be found. The online path determines what can be answered.

Curate, do not hoard

1 CLEAN

Remove duplicates, expired products and conflicting records.

2 STRUCTURE

One product per record; preserve price, stock, skin type and category.

3 CHUNK

Keep each chunk focused enough to retrieve independently.

4 METADATA

Attach metadata for filtering: brand, price, stock, concern, region and freshness.

Tune the shortlist—not the prose

1 TOP-K

How many candidate chunks come back?

2 THRESHOLD

How similar is similar enough?

3 FILTERS

Which price, category, stock or permission rules apply?

4 RERANKING

Which candidates best answer this exact query?

Too few → missing evidence. Too many → noise, higher cost and confused generation.

SESSION 4 · GROUNDING

The answer must stay inside the evidence

SYSTEM INSTRUCTION

Recommend only products present in the retrieved catalogue context.

For each recommendation, cite product name, relevant attributes and price. Explain why it fits the shopper's stated need.

If the catalogue does not support an answer, say what is missing. Never invent a price, stock status, ingredient or policy.

A useful refusal is better than a fluent invention.

SESSION 4 · HANDS-ON

Build it live in Dify

1 Upload

Create a knowledge base and load the catalogue

2 Chunk

Preview boundaries before indexing

3 Embed

Choose a model and create the index

4 Test

Inspect which products are retrieved

5 Wire

Knowledge Retrieval → LLM node

6 Publish

Ask it like a real shopper

Grade the system in the right order

1 CONTEXT RELEVANCE

Did we retrieve the right products?

Fix chunks, metadata, query or ranking.

2 FAITHFULNESS

Is every claim supported by retrieved data?

Fix grounding instructions or evidence.

3 ANSWER RELEVANCE

Did the response solve the shopper's need?

Fix intent handling, output format or prompt.

Watch the retrieved chunks panel—not just the answer.

Name the failure before changing the model

01	Wrong price	→	Metadata / stale data
02	Vague answer	→	Chunks too large
03	Right product absent	→	Retrieval miss
04	Invented policy	→	Grounding failure
05	Good once, bad later	→	No evaluation set

SESSION 4 · CLOSE

A demo works once. A product works repeatedly.

Before production

- A named content owner and refresh schedule
- Permissions enforced before retrieval
- Sources shown with every consequential answer
- A representative test set run after every change
- Known cost per query and a route to a human

SESSION 5 · 90 MIN

How AI Takes Action

Build the vocabulary for connected AI systems—and choose deterministic workflows or flexible agents for the job.

SESSION FLOW

- API + webhook
- Workflow anatomy
- Chatbot vs. agent
- Workflow vs. agent
- Evaluation

HANDS-ON

Map an automation end to end

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API vs. webhook: request versus event

→ API · ASK

Your system sends a request when it wants data or an action.

Example: “Create this CRM lead now.”

! WEBHOOK · TELL

Another system sends an event when something happens.

Example: “A new lead was created.”

An API is a door you knock on. A webhook is a doorbell that rings you.

A workflow connects triggers, rules and actions



If the path is known in advance, encode it. Predictability is a feature.

A chatbot talks. An agent pursues a goal.

CHATBOT

Responds to the message using instructions and context. Usually stops after the answer.

AGENT

Chooses next steps, uses tools, observes results and continues until it reaches a goal or stop condition.

Tool access + decision authority + a loop changes the risk profile.

SESSION 5 · DECISION GUIDE

Workflow or agent?

1 USE A WORKFLOW

Stable sequence · clear rules · low ambiguity · high repeatability · strict audit trail.

2 USE AN AGENT

Variable path · unstructured input · tools must be selected · exceptions require judgement.

3 USE BOTH

Workflow owns controls and approvals; agent handles interpretation inside bounded steps.

Choose the least autonomous system that can reliably do the job.

Did the system do the right thing—not merely say the right thing?

1 TASK SUCCESS

Was the intended outcome achieved?

2 CORRECTNESS

Were actions and outputs accurate?

3 RELIABILITY

Does it work across normal and edge cases?

4 EFFICIENCY

How much time, money and human effort does it require?

5 SAFETY

Did permissions, approvals and limits hold?

Evaluate the full path: trigger → decision → action → side effect → recovery.

SESSION 5 · HANDS-ON

Map one business process

- Choose a recurring process with a clear owner and outcome.
- Mark the trigger, inputs, systems, decisions, actions and approvals.
- Identify where an API is called and where a webhook could start the flow.
- Circle ambiguous steps that may benefit from AI.
- Decide: workflow, agent or hybrid—and define three success measures.

SESSION 5 · CLOSE

Where should autonomy stop?

The important design question is not “Can AI do this?”

It is “What can AI decide, what can it execute, and what still requires approval?”

Bring one candidate workflow to Session 6.

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SESSION 6 · 90 MIN

Building AI Agents & Tool Calling

Combine goals, instructions, memory and tools into a bounded agent—and make every action observable and testable.

SESSION FLOW

- Agent anatomy
- Tool calling
- Agent loop
- Memory + controls
- Use cases + build

HANDS-ON

Design and test an agent

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SESSION 6 · ANATOMY

Five ingredients make an agent

1 GOAL

What outcome is it pursuing?

2 INSTRUCTIONS

What rules and boundaries apply?

3 MODEL

How does it interpret and choose?

4 TOOLS

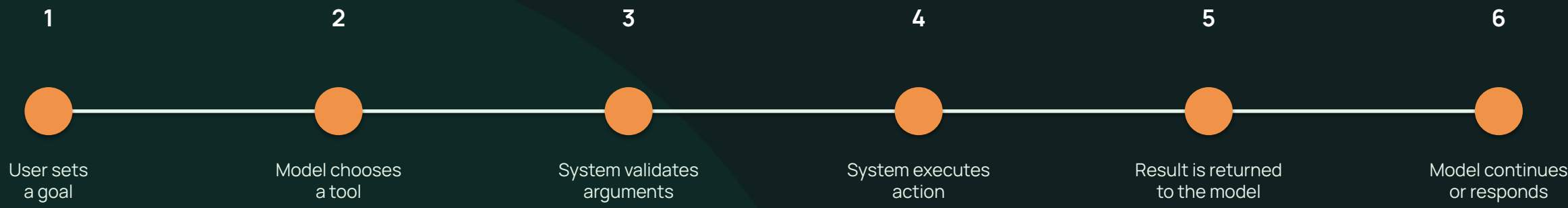
What can it read or change?

5 MEMORY

What should persist across steps?

Without tools, it advises. With tools, it can change the world.

The model chooses a function; the system executes it



Critical boundary: the model proposes. Your application authorises and executes.

SESSION 6 · THE LOOP

Reason → Choose Tool → Act → Observe → Respond

1 REASON

What is the next best step?

→

2 CHOOSE

Which allowed tool fits?

→

3 ACT

Call it with valid inputs

→

4 OBSERVE

Interpret result or error

→

5 RESPOND

Finish—or loop again

Every loop needs a stop condition, retry limit and escalation path.

Remember only what improves the next decision

1 WORKING MEMORY

Current conversation, intermediate results and active plan.

2 LONG-TERM MEMORY

Stable preferences or facts explicitly approved for reuse.

3 SYSTEM OF RECORD

Authoritative business state—the CRM, ticketing or database—not model memory.

Memory creates privacy, freshness and deletion obligations. Store less by default.

Bound the agent before you scale it

- Allow-list tools and validate every argument.
- Use least-privilege credentials and per-user permissions.
- Require approval for irreversible or high-impact actions.
- Set budget, time, retry and loop limits.
- Log tool calls, results, failures and final outcomes.
- Provide a safe fallback and a path for human escalation.

Agents earn value where the path varies

1 RESEARCH

Search sources, compare evidence, create a cited brief.

2 CUSTOMER OPS

Inspect account context, propose resolution, execute approved actions.

3 SALES OPS

Research account, enrich CRM, draft outreach, schedule follow-up.

4 IT SUPPORT

Diagnose issue, consult runbook, run safe checks, open or update ticket.

5 ANALYTICS

Query data, investigate anomalies, build a narrative and recommended next steps.

6 PROCUREMENT

Compare requirements, retrieve the relevant policy, draft evaluation and flag exceptions.

SESSION 6 · HANDS-ON

Design an agent on one page

1 GOAL

One measurable outcome

2 TOOLS

Read vs. write access

3 INSTRUCTIONS

Rules + refusal conditions

4 MEMORY

What persists—and why

5 APPROVALS

Actions needing a human

6 EVALUATION

Test cases + success metrics

SESSION 6 · FINAL CHALLENGE

Workflow, RAG or agent?

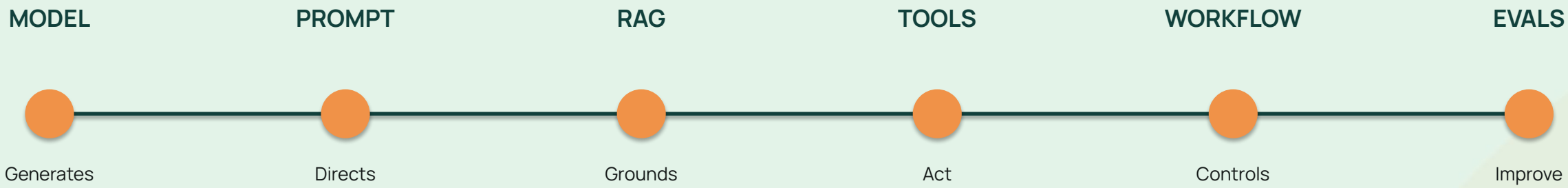
A customer asks for a personalised product recommendation, wants live stock checked, requests a return and asks for the refund to be initiated.

1. What should retrieve evidence?
2. What should follow fixed rules?
3. What may choose tools?
4. Where must a human approve?

Capstone output: architecture + agent card + five evaluation cases.

THE COMPLETE SYSTEM

Model + prompt + knowledge + tools + controls



Capability comes from the model. Reliability comes from the system around it.

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CLOSING

Six sessions later: build with intent

- Understand what the model is doing—and what it is not.
- Brief it with enough context to succeed.
- Ground consequential answers in approved evidence.
- Use workflows for known paths and agents for bounded ambiguity.
- Evaluate outcomes, not demos.

The goal is not more AI. It is better systems.